DATE: PAGE:

SQL Functions

Numeric	character	String	Date
MOD()	LENGTH()	TRIM()	CURRENT_DATE()
POWER()	ASCII()		SYSDATE()
ABS()	UPPER()	Aggregate	NOW()
SQRT()	LOWER()	MAX()	YEAR()
FLOOR()	SUBSTR()	MIN()	
CELL()	REPLACE()	SUM()	conversion/Date
ROUND()	INSERT()	AVG()	DATE_FORMAT()
	REVERSE()	COUNT()	DATE_ADD()
		COUNT(DISTINCT	DATE_SUB()

Numeric functions

-> MOD(): returns the remainder after division

```
SELECT sal, MOD(sal,3) FROM emp;
```

-> POWER(): returns a number raised to a specified power

```
SELECT POWER(2,4);
```

-> ABS(): Returns the absolute value of a number, removing the negative sign

```
SELECT ABS(-15);
```

Scanned with OKEN Scanner

DATE: PAGE:

-> SQRT(): Returns the square root of a no.
SELECT SQRT (16);

-> FLOOR(): ~~Returns the largest integer~~
Rounds a number down to the nearest integer SELECT FLOOR (number);

-> CELL (): Rounds a number up to the nearest integer
SELECT CELL (number);

-> ROUND(): Rounds a number to the nearest integer when no decimal position is specified
SELECT ROUND (5.6); o/p: 6

-> CHARACTER / String functions

-> LENGTH (): returns the no. of characters in a string
SELECT ename, LENGTH (ename) FROM emp;

-> ASCII (): returns the ASCII value of the first characters
SELECT ASCII ('Ammu');

Scanned with OKEN Scanner

DATE: PAGE:

Since the first character is A:
ASCII ('A') = 65

→ UPPER (): CONVERTS text into uppercase
SELECT UPPER ("asHWin");

→ LOWER (): converts text into lowercase
SELECT LOWER ('asHWIN')

→ SUBSTR(): Extracts a part of a string
Ex: SELECT ename, SUBSTR(ename, 2, 8) FROM emp;
Sy: SUBSTR (string, starting_position, num_of_chars)

→ REPLACE (): replace one character or piece of text with another
SELECT REPLACE ('Ashwini', 'S', 'a')

→ REVERSE (): reverses the characters in a string
SELECT REVERSE ('TOM Jerry');
o/p : yrreJ MOT

→ INSTR() : returns the position of a substring char inside a string
SELECT INSTR ('TOM & JERRY', 'J');
o/p : 5

Scanned with OKEN Scanner

DATE: PAGE:

AGGREGATE FUNCTIONS

This function performs calculations on multiple row and returns a single results

-> MAX(): finds the highest value

```
SELECT Max (sal) FROM emp;
```

-> MIN(): finds the smallest salary

```
SELECT MIN(sal) FROM emp;
```

-> SUM(): calculates the total of numeric values

```
SELECT SUM (Sal) FROM emp;
```

calculates total salary

-> AVG(): calculates the average value

```
SELECT AVG (sal) FROM emp;
```

-> COUNT(): counts the no. of row values

```
SELECT COUNT (*) FROM emp;
```

Counts the total no. of rows in emap

-> COUNT (DISTINCT): counts only unique values

```
SELECT COUNT (DISTINCT deptno) FROM Emp;
```

counts how many different departments exists

Scanned with OKEN Scanner

DATE: PAGE:

CONVERSION / DATE FUNCTIONS

-> Date_FORMAT(): converts a date into specified format.

```
SELECT hiredate
DATE_FORMAT (hiredate, '%m-%d-%y')
FROM emp;
```

-> DATE_ADD(): adds a specified amt of time to a date

```
SELECT DATE_ADD (current_date(),
INTERVA 100 DAY);
```

-> DATE_SUB(): Subtracts a specified amt of time from a date

```
SELECT DATE_SUB( CURDATE(), INTERVAL 100 DAY);
```

-> STR_TO_DATE(): converts a text into a date according to specified format

```
SELERT STR_TO_DATE ('30-04-2026', '%d-%m-%y');
```

Scanned with OKEN Scanner

DATE: PAGE:

DATE FUNCTION

- CURRENT_DATE() : returns current date

```
SELECT CURRENT_DATE();
```

- SYSDATE(): returns the current date and time according to the database system clock

```
SELECT SYSDATE();
```

- NOW(): returns current date & time

```
SELECT NOW();
```

- YEAR(): Extracts the year from a date

```
SELECT YEAR (NOW());
```

Scanned with OKEN Scanner